

CSE-4632

Digital Signal Processing Lab

Lab 5

Lab Group: **1**

Thursday, 22th July, 2026

Task 1

The given audio file Lab 4.wav is a recording of a tuning fork ringing. Tuning forks have a particular frequency. Plot the signal from the given audio file. Then, do spectral analysis on the signal to find out what was the frequency of the tuning fork. For, spectral analysis use numpy.fft.rfft function. One way to read the audio file is as follows -

```
from scipy.io import wavfile
# --- Read the audio file ---
fs, data = wavfile.read('Lab 4.wav')
```

Task 2: Forward DFT From Scratch

Write the two following functions, implementing DFT and Inverse DFT. You should not use high level library functions. Implement it from scratch using simple libraries such as numpy, math. The algorithms can be found in Tables 8-1 (Page 154) and 8-2 (Page 160) a. function [ReX,ImX] = myDFT(x) b. function x = myIDFT(Rex,Imx) Verify your DFT and IDFT functions using the example in Figure 8-6 (Page 155)